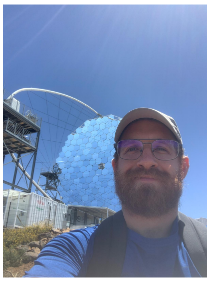

Alessandro Armando Vigliano

+39 331 5875712

viglianoalessandro@gmail.com,
Alessandro.Armando.Vigliano@ts.infn.it

Google Scholar
ORCID
Researchgate profile

RESEARCH FOCUS

Astrophysicist working on multi-messenger and high-energy transients as probes of fundamental physics, with a focus on gamma-ray bursts, multi-instrument data analysis, and photon propagation effects. My research combines observational strategy design, Monte Carlo simulations, automated analysis pipelines, and physical interpretation across X-ray and gamma-ray energies, using data from both space- and ground-based facilities. I am particularly interested in exploiting transient phenomena to probe fundamental physics and cosmology through reproducible, statistically robust methodologies.

APPOINTMENTS

12/2024 – CURRENT

Postdoctoral Research Associate

Research within the Fermi, MAGIC, and CTAO collaborations.
Faculty of Physics, University of Trieste
Project: *"Spectral analysis of Gamma Ray Bursts and Solar Flares emitting in Gamma Rays"*
Responsible: Francesco Longo

EDUCATION

10/2021 – 10/2024

PhD in Mathematics and Physics

Faculty of Mathematical and Physical Sciences, University of Udine
Supervisor: Barbara De Lotto
Co-Supervisors: Francesco Longo, Stefano Ansoldi, Andrea Vacchi
Thesis: ***Novel strategies for the observations of fast gamma-ray transients***

2017 – 2021

Master's degree in Physics

Faculty of Physics, University of Trieste
Curriculum: Astrophysics and Cosmology
Thesis: ***Gamma Ray Burst polarization studies with AMEGO***
Supervisor: Stefano Ansoldi
Co-Supervisor: Francesco Longo

2014 – 2017

Bachelors's degree in Physics

Faculty of Physics, University of Trieste
Thesis: ***Estimation of the contribution of AGN host galaxies to reionization.***
Original title: *Stima del contributo di galassie ospitanti AGN alla reionizzazione.*
Supervisor: Pierluigi Monaco
Co-Supervisor: Fabio Fontanot

INTERNSHIPS

06/2020 – 09/2020

Megalib for simulation of polarized MeV photons: Master Internship

INFN, section of Trieste

02/07/2018 – 07/07/2018

Master Internship at Osservatorio Astrofisico e Astronomico di Asiago

Trieste University

01/09/2016 – 15/02/2017 **Satellite constellation system designing: Bachelor Internship**
PicoSaTs
https://picosats.eu/

SCHOOLS

08/10/2023 – 14/10/2023 **5th MAGIC Hardware School 2023**
MAGIC collaboration, La Palma, Canary Islands

12/02/2023 – 21/02/2023 **Winter School of Theoretical Physics and third COST Action CA18108 Training School ”Gravity – Classical, Quantum and Phenomenology”**
COST Action CA 18108 QGMM, University of Wroclaw, Poland

03/09/2022 – 10/09/2022 **Quantum gravity phenomenology in the multi-messenger approach - COST CA18108 Second Training School**
COST Action CA18108 , Faculty of Physics, Belgrade (Serbia)

28/03/2022 – 08/04/2022 **Astrophysical sources of cosmic rays - ISAPP School Paris**
Institut Pascal, Université Paris Saclay

01/02/2021 – 05/02/2021 **SIGRAV International School 2021: Gravity of Compact Astrophysical Objects and Gravitational Waves**
SIGRAV - Società Italiana di Relatività Generale e Fisica della Gravitazione

15/07/2019 – 26/07/2019 **ICTP School on Geometry and Gravity (smr 3311)**
ICTP - International Centre for Theoretical Physics

11/05/2018 – 18/05/2018 **Trieste Junior Quantum Days**

WORKSHOPS

10/05/2024 **Workshop ”As for the CAT - models and interpretations of Quantum Mechanics”**
University of Trieste, Trieste

01/06/2023 **Workshop ”Simulating ISLE”**
University of Trieste, Trieste

03/04/2023 – 07/04/2023 **Workshop: Machine learning enabled searches for new physics in astrophysical data.**
IFPU, Trieste

09/01/2023 – 13/01/2023 **Workshop ”High redshift Gamma-Ray-Bursts in the JWST era”**
Sexten Center for Astrophysics

18/07/2022 – 22/07/2022 **Workshop: Hands On the Extreme Universe with High Energy Gamma Rays**
Sexten Center for Astrophysics

23/05/2022 – 26/05/2022 **Gravitational Wave Open data Workshop 2022**
LIGO-Virgo-KAGRA, Trieste University

01/02/2022 – 03/02/2022 **International workshop on VHE Data Analysis using Open Source Packages**
University of Kashmir

31/01/2022 **Workshop: ”La formula per la scienza e le sue storie”, science communication**
Trieste University, Parole Ostili

29/03/2021 – 02/04/2021 **Beyond Standard Model: From Theory to Experiment (BSM-2021), Workshop**

Center for Fundamental Physics (CFP), Zewail City of Science and Technology and Sabanci University

14/10/2019 – 16/10/2019 **ICTP Workshop Challenges and Opportunities of High Frequency Gravitational Wave Detection | (smr 3493)**

ICTP - International Center for Theoretical Physics

TALKS, SEMINARS AND CONTRIBUTIONS

24/03/2026 – 26/03/2026 **Theseus Conference 2026**

Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Malaga, Spain

19/01/2026 – 23/01/2026 **Gamma-Ray Bursts: the Great Gig in the Sky**

Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Sexten, Dolomites, Italy

17/11/2025 – 21/11/2025 **2nd LST+MAGIC F2F Science Meeting**

Contributed Talk: *The Soccer Ball Tessellation for the follow-up of Transient Events*
Granada, Spain

30/06/2025 – 04/07/2025 **Advances in Modeling High-Energy Astrophysical Sources: Insights from recent multimessenger discoveries**

Contributed Talk: *Insight on GRB physics from a novel data-driven method for systematic analysis of X-ray light-curves*
Sexten, Dolomites, Italy

24/03/2025 – 28/03/2025 **Celebrating 20 years of Swift Discoveries**

Presented a poster: *Insight on GRB physics from a novel data driven method for systematic analysis of X-ray light-curves*
Firenze, Italy

15/01/2024 – 17/01/2024 **1st VHEGAM meeting: LST proposals by the Italian community**

Contributed Talk: Alternative tiling strategies for transient events
LST collaboration, Bologna, Italy

26/07/2023 – 03/08/2023 **38th International Cosmic Ray Conference (ICRC2023)**

Presented a poster and proceeding on *Gamma Ray Burst localization with GECCO's BGO anticoincidence shields*
Institute for Cosmic Ray Research, University of Tokyo, Japan

19/06/2023 – 23/06/2023 **ASAPP2023 - Advances in Space AstroParticle Physics. Frontier technologies for particle measurements in space. International Conference**

Contributed Talk: *Gamma Ray Burst localization with GECCO's BGO anticoincidence shields*
Agenzia Spaziale Italiana (ASI), Istituto Nazionale di Fisica Nucleare (INFN), Perugia

RESEARCH PROJECTS

01/2024 – ONGOING

Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows

Collaborators: Francesco Longo, Zeljka Bosjnak

Gamma-ray bursts (GRBs) exhibit a rich variety of X-ray lightcurve behaviors, including flares and plateau/shallow decay phases, whose origins remain debated. Existing studies often rely on diverse analysis techniques applied to limited GRB samples, leading to results that may be difficult to generalize. In this study, we introduce a new data-driven, model-independent method for automatically analyzing the Swift-XRT dataset. This approach enables a consistent and comprehensive characterization of GRB afterglow X-ray lightcurves, covering an order of magnitude more events than previous studies. By providing a unified framework for large-scale analysis, this method opens new avenues for identifying robust trends and understanding the physical processes shaping GRB lightcurve evolution.

01/2024 – ONGOING

Development of the divergent mode observation for CTAO

Collaborators: Francesco Longo & INFN Divergent Pointing team

Development of a new divergent pointing mode for future generations of IACTs, particularly CTAO. Comparison with modern tessellation strategies and optimization of divergent pointing strategies for transient follow-up and extragalactic surveys.

11/2021 – 05/2025

New tessellation strategies for multi-messenger observations of VHE fast transients with IACTs, PhD project

Collaborators: Francesco Longo

PhD Thesis

Exploring new pointing strategies for Imaging Air Cherenkov Telescopes (IACTs). I developed a novel strategy for observing very high energy fast transient events, such as Gamma- Ray Bursts (GRBs) and Gravitational Waves (GWs), using IACTs. IACTs offer high angular and energy resolutions at the price of having narrow field of views (FoV), with the current generations of instruments having diameters less than 5 deg. This is a problem for the observations of poorly localised transient events (such as GRBs and GWs) as the localization error regions provided by various facilities have dimensions that go from tens to thousands of square degrees. It is therefore necessary to search the precise location of the source inside these huge error areas with successive observations and given the transient nature of these phenomena the minimisation of the number of pointings is crucial. My approach, termed the "soccer ball tessellation" strategy, leverages a new tessellation for discretising the surface of the sphere in an arbitrary number of equal area hexagonal tiles based on Goldberg polyhedra. This tessellation optimizes the coverage of extended error regions minimizing the number of pointings required to cover them with respect to other discretization schemes such as HEALPix, thereby enhancing the efficiency of observational campaigns. Additionally, the homogeneity of the distribution and minimal tile count per cell dimension make this grid highly suitable for covering even larger areas of the sky: indeed this strategy is a perfect fit for use in any sky survey, such as for example the CTA extra galactic survey, promising improvements in the observational strategies for IACTs and other instruments.

05/2021 – 08/2023

GRB localization capability for GECCO.

Team: Alessandro Armando Vigliano, Francesco Longo, Israel Martinez, Eric Alan Yates, Alexander Moiseev, and Andreas Zoglauer.

Proceeding, Poster

Member of the research team for the Galactic Explorer with a Coded Aperture Mask Compton Telescope (GECCO). Evaluation of the localization capabilities of Gamma Ray Burst (GRB) signals using heavy scintillation shields (BGO).

Accurate localization of transient sources, which carries critical information about their nature, especially given recent discoveries of the GW and high-energy neutrino events, falls within the potential capabilities of gamma-ray telescopes. Presently, the ability for accurate localization in the MeV photon energy range is limited by measurements complications. The Galactic Explorer with a Coded Aperture Mask Compton Telescope ("GECCO") will provide such capabilities thanks to the innovative use of a deployable coded aperture mask in combination with a Compton telescope mode. The spatial arrangement of GECCO's thick and efficient BGO anticoincidence panels can be used to quickly estimate the localization of transient events, allowing a prompt slew of the telescope. GECCO has 8 heavy-scintillator shield panels, arranged on the instrument sides. The ratios between the counts recorded by the panels depends on the direction of the signal with respect to the telescope axis. Simulations were conducted to assess the localization capabilities of the BGO shields, simulating the detection of Gamma Ray Burst signals from the Fermi-GBM catalog with various spectral shapes and time profiles, in order to assess the evolution of their localization error radius with time. The results show how the GECCO shields can achieve localization radii of a few degrees in a short time.

04/2020 – 05/2021

Gamma Ray Burst polarization studies with AMEGO, Master's thesis.

Supervisor: Stefano Ansoldi

Co-Supervisor: Francesco Longo

Master Thesis

Master's thesis on "Gamma Ray Burst polarization studies with AMEGO", with professors Stefano Ansoldi and Francesco Longo, within the context of the AMEGO collaboration. In it, after a thorough study of the mathematical modeling of the production mechanism of gamma ray burst and the different polarization degree predicted by each model, I researched the most recent developments in high energy polarimetry in astrophysics and, in particular, the would-be capabilities of the AMEGO instrument to detect polarization of middle energy gamma-rays (0.2 MeV - 6 GeV). I reconstructed a new modulation description of the instrument that better fits the simulation data I had obtained, which highlights the change of regime from Compton scattering to pair-production in signal detection and which will allow to better model the response of instruments in the latter regime, which is not yet completely clear. To this end I developed a new file for the simulators to replace the standard AMEGO modulation file, which fits better with, and generates more accurate results in, the simulator's environment. Using this novel calibration, I simulated the measurement of different sources. In particular, I focused on gamma ray bursts, to try to establish whether AMEGO will be able to distinguish between the different degrees of polarization of the signal predicted by competing models for the production of such events (such as synchrotron radiation from different magnetic field geometries, Compton drag and photospheric emission), thereby providing important information about the physics of such objects. Such analysis fits perfectly into today's context of nascent multi-channel astronomy, and will allow us to obtain important information about the physics and phenomenology of these objects. Polarization studies are extremely useful in every sector of physics, not just for the observations in the electromagnetic field, as the mathematical description is totally independent from the nature of the waves that are described. This makes such studies particularly interesting for the years to come, with the advancement of instrumentation and the growing possibility of observing this degree of freedom of emissions in multiple energies and channels.

10/2019 – 12/2019

Comparison between solar sails and ion propulsion (powered by solar energy).

Supervisor: Anna Gregorio

As a complement to the Astrophysics Laboratory classes, I proposed and carried out as a project a comparison between two important propulsion systems used in space: solar sails and ion propulsion (powered by solar energy). This required, in addition to an in-depth study of the two systems, the development of some modifications to the equations generally used to estimate the efficiency of rocket propulsion, in order to make a comparison with solar sails.

02/2017 – 07/2017

Estimation of the contribution of AGN host galaxies to Reionization; Undergraduate thesis.

Supervisor: Fabio Fontanot

Co-Supervisor: Pierluigi Monaco

Bachelor Thesis

In my undergraduate thesis on ”Estimation of the contribution of AGN host galaxies to Reionization” with professors Fabio Fontanot and Pierluigi Monaco, I studied the relative contribution of AGN host galaxies to the photon budget needed to end the Reionization epoch. As it is well documented in the literature that in order for galaxies alone to produce the entire necessary photon budget, they should have an extremely high escape fraction (hypothesis not supported by observational data), the contribution of AGNs becomes fundamental. These objects in fact compensate for their low number with a much greater production of ionizing photons than any galaxy and with a particularly high escape fraction. The purpose of my research was to test a corollary of this scenario, namely the hypothesis that, in the event that the galaxy hosting an AGN is also a star forming object, the ionizing photons of stellar origin can benefit from the same (high) escape fraction of the central AGN. To do this I estimated the number of such galaxies using an - at the time - novel combination of many results in order to obtain their UV luminosity. To this end I used the most recent estimates of AGN's luminosity function, the Eddington luminosity to get the central engine mass, the black hole-bulge mass relation, and star forming galaxies' main sequence fitting.

11/2016 – 02/2017

Designed a satellite constellation system to monitor the safety and health of maritime systems.

Supervisor: Anna Gregorio

As a research assignment for my undergraduate internship, I designed a satellite constellation system to monitor the safety and health of maritime systems, under the supervision of Prof. Anna Gregorio (Coordinator of Euclid Instrument Operation Teams of ESA/ESTEC Euclid Mission). When I started mapping the satellites' network, the challenge appeared to be the large number of satellites required to cover all sea routes at all times.Then, exploiting an intuition based on my knowledge as a sailor, I combined the preliminary designs of a constellation of ESA satellites with an in-depth study of the main trade routes, coming to hypothesize a distribution of satellites that guaranteed a continuous coverage of such routes, keeping the number of satellites unchanged, at the expense of a slight loss of coverage around the poles.

RESPONSIBILITIES

20/06/2024 – CURRENT

Gravitational Wave Expert on Call for the MAGIC Collaboration

La Palma, Canary islands

15/03/2024 – CURRENT

Member of the Board of Directors for the Astronomical Observatory ”Helmut Ullric”

Società Astronomica Cortina, Col Drusciè, Cortina d'Ampezzo (Belluno), Italy

15/01/2024 – CURRENT

Burst Advocate for the CTAO-LST1 Collaboration

La Palma, Canary islands

15/01/2024 – CURRENT

Burst Advocate for the MAGIC Collaboration

La Palma, Canary islands

13/01/2025 – 12/02/2025 **Data taking shift on MAGIC**
Operator
La Palma, Canary islands

29/01/2024 – 19/02/2024 **Data taking shift on CTAO-LST1**
Deputy Shift Leader
La Palma, Canary islands

04/08/2023 – 25/08/2023 **Data taking shift on CTAO-LST1**
Operator
La Palma, Canary islands

REFEREEING AND REVIEWING ACTIVITIES

2026 – CURRENT **Referee for Research in Astronomy and Astrophysics (RAA)**

2024 – CURRENT **Referee for Physical Review D (APS)**

2025 – CURRENT **Internal reviewer for the MAGIC Collaboration**

COLLABORATIONS

01/02/2025 – CURRENT **Member of the research team for the Crystal Eye proposal**
Evaluation of the localization capabilities of Gamma Ray Burst (GRB) signals

15/12/2024 – CURRENT **Member & co-proposer of the COST Action CA23130: Bridging high and low energies in search of quantum gravity (BridgeQG)**
Working Group 1: High-energy quantum gravity theory
Working Group 2: High-energy quantum gravity experiment

01/07/2023 – CURRENT **Member of the MAGIC Collaboration**
La Palma, Canary islands

01/11/2021 – CURRENT **Member of the CTAO-LST1 Collaboration**
La Palma, Canary islands

01/01/2020 – CURRENT **Member of the Astronomical Observatory ”Helmut Ullric”**
Società Astronomica Cortina, Col Drusciè, Cortina d'Ampezzo (Belluno), Italy

01/08/2022 – 13/09/2023 **Member of the COST Action CA18108: QG-MM ”Quantum gravity phenomenology in the multi-messenger approach”**
Working Group 3: Gamma rays

20/05/2021 – 30/09/2024 **Member of the research team for the Galactic Explorer with a Coded Aperture Mask Compton Telescope (GECCO).**
Evaluation of the localization capabilities of Gamma Ray Burst (GRB) signals using anti-coincidence shields (BGO)

TEACHING

2025 **Co-Supervisor for bachelor degree student thesis.**
”Analysis of AGN VHE emissions with the MAGIC telescopes”
University of Trieste, Trieste

2024 **Co-Supervisor for master degree student thesis.**
”Searching for high-z GRBs in Fermi-LAT data using ML techniques.”
University of Trieste, Trieste

2023 – 2024 **External referee (INFN) for master degree student internships.**
5 students: MC simulations with Geant4 for medical physics
1 student: simulations of HE astrophysical signals, multiwavelength data analysis, CNN
University of Trieste, Trieste

02/10/2023 – 15/12/2023 **Teaching assistant for General Physics to the Mathematics curriculum**
University of Udine, Udine

27/02/2023 – 09/06/2023 **Teaching assistant for Analysis II to the Mathematics curriculum**
University of Udine, Udine

OUTREACH AND PUBLIC ENGAGEMENT

02/2026 **Lecture at high school**
Invited lecture on astrophysics and astronomy to high school students
Padova, Italy

09/2025 **German Italian Physics Exchange (GIPE)**
Presentation on gamma ray astronomy
University of Trieste, Italy

2022 – 2025 **Studiare Fisica a Trieste**
Presentation on gamma ray astronomy for incoming student orientation
University of Trieste, Italy

20/05/2025 **Pint of Science**
Public talk on gamma ray astrophysics: *Il Cielo nel Pallone*
Trieste, INFN & INAF, Italy

25/02/2025 – 26/02/2025 **2025 Italian Inter-regional Astronomy Olympics, judge and jury member.**
INAF, Italy

2021 – 2023 **Involved in the organization/staff of the joint INFN-INAF stand at the Trieste NEXT fair.**
University of Trieste, Italy

24/02/2022 – 26/02/2022 **2022 Italian Inter-regional Astronomy Olympics, judge and jury member.**
INAF, Italy

05/11/2021 – 13/11/2021 **International Remote Astronomy Olympics (IRAO 2021), Invigilator**
INAF, Italy

OTHER RESPONSIBILITIES

01/01/2021 – 29/09/2022 **Director of the competitive team and of relations with the national bodies CSEN and FIJLKAM. Representative of the athletes on the board of directors.**
Unione Judo Trieste

01/11/2016 – 28/11/2016 **First Sempai Seminar (Takemusu Aikido), assistant and organizator.**
Società Ginnastica Triestina

13/07/2015 – 27/07/2015 **Sailing Instructor**
Scuola Vela Golfo di Trieste, Muggia, Italy

SKILLS

LANGUAGES

Italian – mother tongue
English – C1
- IELTS band 8 (23-24/07/2021)

Spanish – B1

DIGITAL SKILLS

Programming languages
- **Python**
- Wolfram Mathematica
- Fortran (77, 90, 95)
- C
- Linux Bash
- IDL
- Arduino IDE

MEGAlib
AGI System Tool Kit (STK)
Git

OTHER CERTIFICATIONS

BLSD - Basic Life Support and Defibrillation — Basic Life Support with a Defibrillator, 23/10/2025

BLS - Basic Life Support — Basic Life Support, 03/06/2025

Nautical license — Nautical license, 04/07/2023, n° 953520

Scuba Diving — PADI Open Water Diver, 24/08/2021, n° 2109EI8387

HOBBIES AND INTERESTS

Competitive Judo, BJJ/Grappling, MMA

I now train everyday in the Mixed Martial Arts (MMA) competitive team of the Bear Brothers Club in Trieste. I trained in the club's competitive judo team (Unione Judo Trieste), regularly participating in Judo, BJJ and Grappling competitions. This has taught me to be organized and disciplined. From January 2021 to September 2022 I have also held the role of director of the competitive team and its relations with the national body CSEN.
Some achievements:
- 1° Kyu Brown Belt Judo
- 2019, 43° International Trophy Judo, Pordenone, third place (with one year of experience)
- 2018, Italian National Grappling Championship, C series, third place (with few months of experience)
- 2019, Fight Gala, Grappling, Lignano, first place
- 2019, Portogruaro Trophy, Grappling, first place

I practiced Aikido (Società Ginnastica Triestina) during my undergraduate studies, taking an active role in the organization of international technical seminars with masters from all over the world (Italy, France, Germany and Hungary).

Sailing, Foiling
- International Moth: Italian Nationals 2013 and 2014, Europeans 2013
- FIV Assistant Instructor , May 2016
- CVC (Centro Velico Caprera): All courses
- Muggia, Porto san Rocco, Scuola Vela Golfo di Trieste

Free Climbing and Bouldering
- Trained with the Scoiattoli team in Cortina d'Ampezzo, pupil of Massimo Da Pozzo.
- Trained with the 360 team in Cortina d'Ampezzo, max level 7a

Skiing
- giant slalom and freestyle
- a little of freeride
- Trained with the Scuola Azzurra in Cortina d'Ampezzo

Music I play different instruments at different levels, in particular flute, Irish tin whistle, guitar, harmonica and traditional Apulian tambourine (pizzica from Salento)

Brain Teasers I love challenging puzzles and brain teasers. After solving the Rubik's cube without the appropriate algorithms (developing my personal set of moves) I now enjoy speedcubing (personal record on the 3x3: 23s).

First/contact author & main contributor

[1] A. A. Vigliano and F. Longo. *Soccer Ball Tessellation: a novel method for tiling large uncertainty areas with narrow field of view telescopes*. In preparation. 2026.

Article based on the results of the PhD thesis and its new developments

[2] A. A. Vigliano, F. Longo, and Z. Bosnjak. *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*. In review A&A. 2026.

Conceived the analysis, developed the methodology, performed the data analysis, wrote the paper

[3] A. A. Vigliano. "Novel strategies for the observations of fast gamma-ray transients". Publicly available online. PhD thesis. Università degli Studi di Udine, Udine, Italy, 2025.

PhD thesis. I conceived the analysis, developed the methodology, performed the data analysis, wrote the paper

[4] A. A. Vigliano and F. Longo. "Gamma-ray Bursts: 50 Years and Counting!" In: *Universe* 10.2 (2024), p. 57. DOI: 10.3390/universe10020057.

A review of the history of gamma-ray burst observations and studies. I conducted in-depth historical and scientific research and wrote the manuscript.

[5] A. Kouchner, A. A. Vigliano, and M. Lamoureux. "Detection and phenomenology of cosmic neutrinos". In: *PoS* QG-MMSchools (2024), p. 005. DOI: 10.22323/1.440.0005.

A review of the history of observations and studies of astrophysical neutrinos as part of the Second School of the COST Action QGMM. I conducted extensive historical and scientific research and wrote the manuscript.

[6] A. A. Vigliano et al. "Gamma Ray Burst localization with GECCO's BGO anti-coincidence shields". In: *PoS* ICRC2023 (2023), p. 748. DOI: 10.22323/1.444.0748.

Conceived the analysis, developed the methodology, performed the data analysis, wrote the paper

Contributor

[7] R. Aloisio et al. "Crystal Eye: All sky MeV monitor with high precision real-time localization". In: *Astroparticle Physics* 174 (2026), p. 103171. ISSN: 0927-6505. DOI: https://doi.org/10.1016/j.astropartphys.2025.103171. URL: https://www.sciencedirect.com/science/article/pii/S0927650525000945.

Contributed to the evaluation of the localization capabilities of transient events (especially GRBs) through classical and ML techniques

[8] R. Alves Batista et al. "White paper and roadmap for quantum gravity phenomenology in the multi-messenger era". In: *Class. Quant. Grav.* 42.3 (2025), p. 032001. DOI: 10.1088/1361-6382/ad605a. arXiv: 2312.00409 [gr-qc].

Contributed to writing and review. Active member of the COST Action QGMM and co-proposer of its followup COST Action BridgeQG

[9] A. Campoy-Ordaz et al. "Search for Lorentz Invariance Violation with time-lag on gamma-ray Cherenkov Telescope data : From the data to the time lag constraints". In: *PoS* QG-MMSchools (2024), p. 009. DOI: 10.22323/1.440.0009.

Contributed to the writing (tutorial section of the paper)

[10] A. Coogan et al. "Hunting for dark matter and new physics with GECCO". In: *Phys. Rev. D* 107.2 (2023), p. 023022. DOI: 10.1103/PhysRevD.107.023022. arXiv: 2101.10370 [astro-ph.HE].

Contributed with the evaluation of the detector's capabilities

[11] A. Donini et al. "Performance study update of observations in divergent mode for the Cherenkov Telescope Array". In: *PoS* ICRC2023 (2023), p. 840. DOI: 10.22323/1.444.0840. arXiv: 2309.14106 [astro-ph.HE].

Contributed as an expert on tessellation strategies for the follow-up of poorly localized transient events

The publications listed below are large-scale collaborative works. I am member of both analysis and operations teams contributing to data taking, validation, and internal reviews.

Collaboration papers

[12] S. Abe et al. "MAGIC observations of NGC 4278. The first low-luminosity radio galaxy with compact jets detected at TeV energies". In: (Mar. 2026). arXiv: 2603.22955 [astro-ph.HE].

[13] K. Abe et al. "VHE gamma-ray intranight variability from BL Lacertae during the extreme flaring state of 2022". In: (Mar. 2026). arXiv: 2603.19889 [astro-ph.HE].

[14] K. Abe et al. "First detection of VHE gamma-ray signal from the FSRQ TON 0599". In: *Mon. Not. Roy. Astron. Soc.* 546 (2026), p. 4. DOI: 10.1093/mnras/stag032. arXiv: 2601.03873 [astro-ph.HE].

[15] J. Abhir et al. "Prompt Searches for Very-high-energy γ-Ray Counterparts to IceCube Astrophysical Neutrino Alerts". In: *Astrophys. J.* 997.2 (2026), p. 141. DOI: 10.3847/1538-4357/ae2c4e. arXiv: 2512.16562 [astro-ph.HE].

[16] K. Abe et al. "CTAO LST–1 observations of magnetar SGR 1935+2154: Deep limits on sub-second bursts and persistent tera-electronvolt emission". In: *Astron. Astrophys.* 706 (2026), A25. DOI: 10.1051/0004-6361/202555431. arXiv: 2512.05634 [astro-ph.HE].

[17] K. Abe et al. "VHE γ-ray observations of bright BL Lacs with the Large-Sized Telescope prototype (LST-1) of the CTAO". In: *Mon. Not. Roy. Astron. Soc.* 544.1 (2025), pp. 669–686. DOI: 10.1093/mnras/staf1728. arXiv: 2510.03702 [astro-ph.HE].

[18] M. Tambone et al. "WINK: Advancing X and Gamma Ray Detection Technology for Space Observations". In: *PoS* ICRC2025 (2025), p. 857. DOI: 10.22323/1.501.0857.

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